

AI GENERALIST ACCELERATOR · 14-DAY AI SPRINT

Day Three

Automations From First Principles: LLMs, Agents, and Your First n8n Workflow

A cleaned, detailed record of the live session and breakout briefing

Topics covered: The Next-Word Predictor · Memory, Tools, and Context · LLMs Versus AI Agents · Triggers and Actions · Why n8n, and Three Ways to Run It · The Spam-Filter Build · Models, Credentials, and Costs · Publishing, and the Workbook

About This Session

Day three moved the cohort from asking AI to do things to getting AI to do things automatically, and it did so with a new pairing: Sumedha hosting her first main session, apologising in advance for a physically cracked laptop camera and promising the AI-generated session playlist on graduation day, and Jitesh Dugar teaching, founder at Media Jade, an authorised Zoho partner, a verified n8n creator, and a builder of automation solutions for the past seven years. His opening was a contract, two and a half hours of undivided attention in exchange for an extremely valuable resource, with two riders that shaped the whole evening: he would deliberately make mistakes that the room had to catch, and nobody was to build along, because the hands-on belongs to the breakout and following the screen live means missing the ideas. The session refused to start at n8n at all, spending its first hour building the vocabulary, LLMs, memory, tools, agents, triggers, actions, so that the tool, when it finally appeared, was obvious rather than magical.

HOW TO USE THESE NOTES

The workbook rebuilds the session's exact spam-filter workflow step by step, with a screenshot of the finished flow, a written guide, and, because breakout rooms are never recorded, a full walkthrough video. It also ships a working OpenAI API key valid only until 6 pm IST the next day, strictly for the workbook; after it expires, generate a free Gemini API key or recharge OpenAI with its five-dollar minimum.

Two more resources live on the platform: the exported workflow file Jitesh shared in chat, importable straight into any n8n account, and, under day three's resources, a step-by-step video on self-hosting n8n for anyone who asked how to run it beyond the trial. Bonus office hours debut tomorrow at 6 pm IST, with the main session at 7.30.

The Word Game, and What an LLM Really Is

The teaching began as a game. Round one asked the chat to fill blanks, better late than, twinkle twinkle little, Mary had a little, hope this email finds you, thanks in, and every answer landed identically, which set the question that mattered: nobody supplied the answers, so how did everyone know? Because brains trained on a thousand repetitions predict the next word automatically, and the machine-world name for exactly that ability is the LLM, the large language model, which Jitesh insisted on demystifying as the world's best next-word predictor, nothing more, its guesses not random but the winners of enormous behind-the-scenes calculation, which is why it feels like magic and is not. Round two changed the questions, the best cricketer of all time, the best place to live in Bengaluru, my favourite food, and the room that had marched in lockstep scattered across Bradman, Tendulkar, biryani, and risotto, the point being personalisation and context: prediction depends on what surrounds it, which is why the same model given the same prompt can return two different sales emails, an experience no machine ever gave us before LLMs. The mechanics got a full walkthrough, a question like what is the capital of the United Kingdom is converted to tokens, the training data is consulted, and the answer is typed one predicted token at a time, the, capital, city, of, the, United Kingdom, is, London, with every next word winning a weighted competition, so a four-hundred-word email is not drafted, it is predicted four hundred times in a row. The two objections were met head on: images and videos are tokens too, the same prediction engine at far higher complexity, and code reads as brilliant because the model has trained on so many patterns of good Python that its guesses stopped looking like guesses.

Memory, Tools, and the Meaning of Context

Prediction alone forgets. Ask the capital of the United Kingdom, get London, then ask who is the Prime Minister of this country, and a bare LLM, which has no built-in memory, receives only the second question and must answer, sorry, which country? Memory is the fix, and its mechanism is the session's key definition: with memory, question two is fed to the model together with question one and answer one, and context is everything you mix along with your original question. Feed good context and LLMs do really well, which the room compressed to context is king. But even a remembering LLM only talks; ask it to actually send a drafted email to ten prospects and it must refuse, because it cannot touch the real world. The remedy is tools, described exactly as giving the LLM a hand: access to your inbox so it can send and read email, to your calendar so it can book and cancel meetings, to a food-delivery app so it can genuinely order dinner. Every fancier term collapses into this one, a connector is a tool, and MCP, the model context protocol everyone has heard invoked, is simply a way of giving AI the ability to connect to external systems, tools again, whether wired as APIs or MCPs.

LLMs Versus AI Agents

With the three pieces named, the definition wrote itself: an AI agent is a mix of an LLM to think, a memory to remember, and tools to execute. Then came the question Jitesh had planted at the start, is ChatGPT an LLM, and the room, which had voted yes an hour earlier, flipped, correctly: three years ago ChatGPT effectively was an LLM, writing poems with no memory and no hands, but today it is a full-blown agent, and so are Gemini and Claude as products. The LLMs are the models inside, on Gemini the 3.6 Flash, 3.6 Thinking, and 3.1 Pro switcher, on Claude the Fable 5.1, Opus 5, Sonnet 5, and Haiku 5 list, while the product wrapping them carries the built-in memory and the connectors, which is why Claude, told to read Slack, check the CRM, or deploy an app to Netlify, simply does it. The working distinction to keep is that LLMs act on instructions while agents act on goals, illustrated with a goal nobody scripted step by step: free up my calendar meetings for Friday. The agent checks the calendar, finds meetings with Daniel, Ranjith, and James, emails each asking to reschedule, waits for the replies, moves the meetings of those who agree, then asks itself whether the goal is achieved and loops back into next steps until it is. Balance came with the caveats: LLMs remain the superior choice for the plain jobs, summarising, generating, translating, transforming, and classifying, that last one about to matter, and agents demand deliberate decisions about which of your accounts they may touch.

Triggers, Actions, and the Tools That Build Them

Automation got the same from-scratch treatment through a coffee shop. Place the order and pay, the cashier accepts it, the barista prepares the coffee, your name is called, you collect, you enjoy, and the room correctly picked the step that set everything else in motion: the order. In automation language that step is the trigger, everything downstream of it is the actions, and an automation is nothing but a trigger plus actions, a definition that survives every synonym, workflow, flow, and every platform. The Amazon order whose warehouse machinery runs invisibly to your doorstep is exactly such a system, historically the work of hundreds of developers, and the modern twist is that workflow automation tools now let anyone assemble the same flowcharts visually, block after block, without a line of code: Zapier, Make.com, n8n, Power Automate, Pabbly Connect, Integreatly, and a market of others. The course's choice of n8n was defended before anyone asked, this is not Outskill promoting a product: among all of them n8n stands out for flexibility and for being the only one that is open source, and every concept taught tonight transfers unchanged to the rest, only the interface differs. The name itself is a lesson, n8n abbreviates nodemation, node-based automation, n plus the eight letters of odematio plus n, and node, nothing to do with Node.js, simply means each step, each block, of a workflow, trigger and actions alike. The site tour corrected its own marketing, the advertised five hundred integrations are in fact over twenty-one hundred, Gmail, Outlook, Google Sheets, Teams, Slack, Telegram, OpenAI, and almost certainly whatever CRM you already run, and the privacy question got its honest answer, n8n documents SOC 2, GDPR, and the standard compliances in full on its site.

Three Ways to Run n8n

Because someone always asks what open source means, the pricing section began on GitHub, a Google Drive for code, where n8n has published the entire codebase powering its product, downloadable by anyone as a zip, with the reassurance that public code does not mean public data, your information stays secure on the servers you run. That fact creates three ways to use the tool. The cloud plan, hosted by n8n itself, opens with a fourteen-day free trial and then runs roughly twenty to twenty-four euros a month depending on billing, and it is the recommendation for the course, no setup, everything just works. Self-hosting on your own computer is genuinely free forever, the open-source code running locally on your machine's own memory and space, with one honest catch, it stops the moment the computer sleeps, so nothing runs 24-7. And self-hosting on a VPS, a virtual private server rented from DigitalOcean, Hostinger, AWS, Google Cloud, Railway, or any of a dozen providers, keeps the free software running around the clock for roughly six to ten dollars a month of rent. The second and third routes involve real technical work, which is exactly why the full step-by-step self-hosting video sits in day three's resources on the platform rather than in the session's scarce minutes.

The Problem, and the Flowchart-First Rule

The build began, as Jitesh insisted every build should, with a drawn flowchart rather than an open tool. The scenario: you run an agency selling websites and digital marketing, your site carries an inquiry form, name, company, email, project brief, submit, and leads arrive in the thousands, spam mixed inseparably with gold, so the sales team needs the two filtered into separate buckets at scale. The room solved the design live. The trigger is the submit click. The first action validates the data, the working assumption being simply that the project brief must not be empty, stop there if it is. Then the decisive step, classifying spam versus genuine, where the chat split between keyword matching and an LLM, and keyword matching lost on one counterexample: do you sell umbrellas contains no spam keywords and sails through any list, while an LLM, whose classification strength had been planted an hour earlier, judges meaning. The routing completed the chart, spam inquiries appended to a spam sheet, genuine ones to a genuine sheet, with a deliberate design decision defended against the do-nothing suggestion: LLMs are prediction engines and will occasionally misfile a real lead, and a mislead lost forever costs far more than a spam row, so spam goes to a reviewable sheet, not the void.

The Build: Triggers, the Form, and the First Deliberate Mistake

Inside a fresh cloud workspace, Create Workflow, named Website Spam Checker, the build-with-AI button acknowledged and set aside in favour of learning by hand, the first plus button asked the question every workflow asks, what triggers this? The trigger taxonomy got its full tour. Manual, a human clicks a button. On app event, the richest family, over a thousand apps whose happenings can start a flow, Gmail's on message received, Google Sheets' row added or updated, a WhatsApp message, a meeting landing on your calendar. Schedule, the clock as trigger, illustrated with the automation everyone wanted by the end, every morning at 8 am read the last day's unread email and send an AI summary of the five that matter, runnable by minute, hour, day, week, or month. On webhook call, deferred as too complex for a first lesson with a promise to return. And on form submission, the day's choice, because n8n can host the form itself. The form came together field by field, title Project Inquiry, description dictated through Whisper Flow, your name and company as text inputs, email switched to the email type, and the project brief given a text area for paragraphs, with Execute Step popping the live form for a first test. The test exposed the gap on purpose, a submission with every field blank sailed through, fixed by Add Attributes marking name, company, and email required while the brief stayed deliberately optional to keep the validation step meaningful. Two pieces of workspace literacy rounded it off: the output panel's schema, table, and JSON views are three lenses on identical data, and Pin Data saves a test submission so every later node has something real to execute against.

The validation node introduced the If step and, with it, the evening's first planted error. The condition read project brief is not empty, Execute Step returned true, and the room was invited to find the flaw: the value being checked was typed in as a fixed string, the sample text itself, not the live field, so the check would pass forever regardless of what users submitted. The fix is the single most important mechanic in n8n, drag the field from the input panel on the left into the condition, the cursor becomes a hand, and the moment it drops, the toggle flips itself from fixed to expression, fixed meaning static, expression meaning dynamic, a value that changes with every run. The condition now tested the real submission, true and false branches lit up, and the chat's flood of hundreds confirmed the lesson had landed.

The AI Step: Credentials, Model Ladders, and the Second Mistake

Making the automation an AI automation took one node and three lessons. The node, reached by searching the AI section and choosing OpenAI with the message-a-model action, you are literally telling a model, here is my inquiry, classify it, exactly like talking to ChatGPT but inside a workflow. The first lesson was credentials, defined as what gives n8n the ability to communicate with external apps, and the distinction most of the room had never been taught: OpenAI and ChatGPT are two products of one company, ChatGPT a twenty-dollar-a-month subscription interface, the OpenAI platform a prepaid, pay-per-token system with a five-dollar minimum, and connecting n8n uses the latter, never your ChatGPT plan, a split that applies identically to Claude, Gemini, and DeepSeek. The key itself takes a minute, platform.openai.com, API keys, create new secret key, optionally expiring, copy it once and paste it into a new n8n credential. The second lesson was the model ladder, drawn as brains of increasing size: in the OpenAI world Nano, then Mini, then the general model, then Pro, intelligence and cost rising together, Pro running as much as two hundred times Nano's price while Nano answers fastest, and in the Claude world the same ladder reads Haiku, Sonnet, Opus, Fable. The method: match the brain to the task, start at Mini when unsure and escalate, and since classifying an inquiry is about as simple as tasks get, the room voted for the smallest brain and GPT-5.4 Nano got the job. The prompt was outsourced to Claude on a quick answer, and came back with the whole classifier's policy, spam means SEO, backlink, and guest-post agencies, link builders, freelancers offering services, crypto, loans, and investment pitches, answer yes; genuine means website redesigns, ads, pricing, timelines, partnerships, referrals, answer no; and two lines worth stealing for every classifier ever built, if unsure answer no, because a mislead costs more than a spam message, and short or poorly written messages are not spam by themselves, many genuine leads write briefly. Execute Step returned no, and the second planted mistake got its moment: the room caught that the prompt never contained the inquiry it was meant to judge, fixed by flipping the field to expression and dragging the project brief in under a closing line, project inquiry is below, so every one of a hundred daily submissions rides into the same prompt dynamically. Routing the answer introduced one more node by choice: an If would have worked for yes and no, but Switch, configured with two rules on the model's text output and its outputs renamed Spam and Genuine, exists precisely because it allows ten branches where If allows two, and knowing it exists is the point.

Sheets, Publishing, and the Live Miss

The destinations were built in sixty seconds of Google Sheets, a document named Project Inquiries with two tabs, Spam Inquiries and Genuine Inquiries, each carrying name, company, email, project brief, and submitted at. Connecting them taught the friendlier face of credentials, Google apps offer a sign-in-with-Google button, two or three clicks and a permissions screen, after which n8n can read, write, and create in your sheets. The action chosen was append row, not update, because every inquiry is a new row, the document and tab were picked from n8n's own listing, and the column mapping was pure drag and drop, each form field pulled from the left panel into its column until Execute Step landed a real row in the genuine tab. The spam branch took one right-click, duplicate the node, drag its wire onto the Switch's spam output, flip the tab setting, remap the same five fields, and rename both steps honestly, Add to Genuine Sheet, Add to Spam Sheet. Publishing turned the rehearsal into production, the form node's production URL now a live public page, brandable, embeddable on a real website, its n8n badge removable and its styling customisable, and the first production test delivered the night's best unscripted lesson: a submission reading I'm selling SEO services, textbook spam, landed in the genuine tab. The Nano brain was not up to the job, so the model was switched to GPT-5.4 Mini, the workflow republished, the same inquiry resubmitted, and it filed correctly into spam, no button pressed, everything visible after the fact in the Executions tab that logs every automatic run, a perfect live demonstration of the ladder logic taught an hour earlier, start small, escalate on evidence. The finished workflow was exported and dropped in the chat for everyone to import.

THE MENTAL MODEL

An LLM predicts the next word; add memory and it remembers; add tools and it acts; that mix is an agent, which pursues goals where an LLM follows instructions. An automation is a trigger plus actions, drawn as a flowchart before it is built, and in n8n every block is a node whose values must be dragged in as expressions, never typed as fixed text. Credentials let n8n touch your apps on pay-per-token API billing, not your chat subscription; start with a small-brain model and climb only when it misses.

Closing, the Workbook, and What Comes Next

The confidence poll ran the honest spread, hundreds from some, twenties and forties from others, and Jitesh's closing reassurance was calibrated to a first lesson on a genuinely new tool: forty or fifty percent on day one of n8n is very good, and the remaining half arrives only through practice, which is what the breakout existed to start. Sumedha's wrap pointed everyone at the machinery for exactly that: the workbook rebuilds the identical spam filter with a flow screenshot, a step-by-step guide, and a full video for anyone who missed the unrecorded rooms, and it ships a shared OpenAI API key that must be copied exactly as printed, valid only until 6 pm IST tomorrow, right before the first office hour, and only for the workbook, with free Gemini API keys or a five-dollar OpenAI recharge as the paths afterward. The self-hosting question got its permanent answer, a complete video guide under day three's resources on the platform, and anyone missing their day one or two badges writes to the support address or the tech-support channel. The rooms ran into the late evening, and the sign-off set the new daily rhythm in motion: bonus office hours debut tomorrow at 6 pm IST on Slido, a half-hour break, then day four's main session at 7.30.

Key Takeaways

Stripped to its essentials, the session made the following points.

- **An LLM is the world's best next-word predictor.** Trained on tokens, it types answers one weighted guess at a time, which is why identical prompts can differ, why context changes everything, and why images, video, and even brilliant code are the same prediction engine at higher complexity.
- **Context is everything you mix along with your question.** Bare LLMs have no memory, so follow-ups fail until previous questions and answers ride along, and tools, connectors and MCPs included, are simply the hands that let a model act on email, calendars, and apps instead of only talking.
- **Agents are LLM plus memory plus tools, and they chase goals.** ChatGPT and Claude the products are agents wrapping the Fable, Opus, Sonnet, and Haiku class of models; free up my Friday calendar is a goal an agent loops on until met, while summarise, translate, and classify stay with the plain LLM.
- **Every automation is a trigger plus actions.** The coffee order that sets the barista moving is the trigger, everything after is actions, visual tools like n8n, Zapier, and Make assemble them without code, and the flowchart is always drawn before the tool is opened.
- **n8n is the open-source pick with three ways in.** The hosted cloud plan's fourteen-day trial then roughly twenty euros a month, free self-hosting that sleeps when your laptop does, or a six-to-ten-dollar VPS running 24-7, with the code public on GitHub and your data private throughout.
- **Drag, never type: expressions over fixed values.** The two planted mistakes taught the core mechanics, conditions must test the dragged live field, not a pasted sample, and a classifier prompt is useless until the actual inquiry rides into it dynamically.
- **Know what you are paying for.** n8n credentials use the OpenAI platform's prepaid tokens, never your ChatGPT subscription, and models are a brain ladder, Nano to Pro, Haiku to Fable: start small, escalate on evidence.
- **Publish, then trust the Executions tab.** The production form ran itself end to end, the Nano brain's misfiled spam pitch was caught in the logs and fixed by a switch to Mini, and the exported workflow plus the workbook's API key mean everyone rebuilds it tonight.

End of Day Three notes. Next: bonus office hours at 6 pm IST, and Day Four.